



Top 14 CAT Miscellaneous LR Questions With Video Solutions

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Questions

Instructions

Mathematicians are assigned a number called Erdos number (named after the famous mathematician, Paul Erdos). Only Paul Erdos himself has an Erdos number of zero. Any mathematician who has written a research paper with Erdos has an Erdos number of 1. For other mathematicians, the calculation of his/her Erdos number is illustrated below:

Suppose that a mathematician X has co-authored papers with several other mathematicians. 'From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y . Then X has an Erdos number of $y+1$. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity. :

In a seven day long mini-conference organized in memory of Paul Erdos, a close group of eight mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3. The Erdos numbers of B, D, E, G and H remained unchanged with the writing of this paper. Further, no other co-authorship among any three members would have reduced the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fifth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining six were unchanged with the writing of this paper.
- No other paper was written during the conference.

Question 1

The Erdos number of E at the beginning of the conference was:

- | | |
|----------|---|
| A | 2 |
| B | 5 |
| C | 6 |
| D | 7 |
| E | 8 |

$$\begin{aligned} \text{Erdos} &\rightarrow 0 \\ \text{number} &\rightarrow 1 \\ y &\rightarrow y+1 \\ &\rightarrow \infty \end{aligned}$$

Erdős was a famous Hungarian mathematician, who collaborated with his/her Erdos number is illustrated below:

Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y . Then X has an Erdos number of $y+1$. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

On the third day of the conference, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.

On the fourth day, co-authored a paper with F which reduced the Erdos number by 0.5. The Erdos numbers of the remaining three members unchanged with the writing of this paper.

No other paper was written during the conference.



Question 2

How many participants had the same Erdos number at the beginning of the conference?

- A 2
- B 3
- C 4
- D 5
- E Cannot be determined

Erdos $\rightarrow 0$
math $\rightarrow 1$
 $y \rightarrow y+1$
 $\rightarrow D$

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Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

The seven day long mini-conference organized in memory of Paul Erdos, a close group of mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3.5. Further, no other co-authorship among any three members would reduce the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members remained unchanged with the writing of this paper.
- No other paper was written during the conference.

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Question 3

The Erdos number of C at the end of the conference was:

- A 1
- B 2
- C 3
- D 4
- E 5

Erdos $\rightarrow 0$
math $\rightarrow 1$
 $y \rightarrow y+1$
 $\rightarrow D$

cracku

Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

The seven day long mini-conference organized in memory of Paul Erdos, a close group of mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3.5. Further, no other co-authorship among any three members would reduce the average Erdos number of the group of eight to as low as 3.

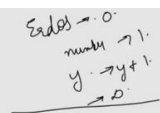
- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members remained unchanged with the writing of this paper.
- No other paper was written during the conference.

VIDEO SOLUTION

Question 4

The person having the largest Erdos number at the end of the conference must have had Erdos number (at that time):

- A 5
- B 7
- C 9
- D 14
- E 15



ERDOS NUMBER: ERDOS NUMBER OF A PERSON IS DEFINED AS THE LENGTH OF THE SHORTEST PATH FROM HIS/HER ERDOS NUMBER IS ILLUSTRATED BELOW:

Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y . Then X has an Erdos number of $y+1$. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

A seven day long mini-conference organized in memory of Paul Erdos, a close friend of many mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3.5. Further, no other co-authorship among any three members would reduce the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members remained unchanged with the writing of this paper.
- No other paper was written during the conference.

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Question 5

How many participants in the conference did not change their Erdos number during the conference?

- A 2
- B 3
- C 4
- D 5
- E Cannot be determined

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Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

On the third day of the conference F co-authored a paper jointly with A and C. The average Erdos number of the group of eight mathematicians to whom A, B, C, D, E, F, G and H discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.

On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members were unchanged with the writing of this paper.

No other paper was written during the conference.

VIDEO SOLUTION

Instructions

Directions for the next 2 questions:

A, B, C are three numbers.

Let $@(A, B)$ = average of A and B,

$/(A, B)$ = product of A and B, and

$X(A, B)$ = the result of dividing A by B

Question 6

Average of A, B and C is given by:

- A $@(/(@/(B, A), 2), C), 3)$
- B $X(@/(@/(B, A), 3), C), 2)$
- C $/(X(@/(B, A), 2), C), 3)$
- D $/(X(@/(@/(B, A), 2), C), 3), 2)$

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Question 7

The sum of A and B is given by:

- A $/(@/(A, B), 2)$
- B $X(@/(A, B), 2)$
- C $@/(A, B), 2)$
- D $@(X(A, B), 2)$

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Instructions

For the following questions answer them individually

Question 8

There were 'x' pigeons and 'y' mynahs in a cage. One fine morning, 'p' birds escaped to freedom. The bird-keeper, knowing only that $p = 7$, was able to figure out without looking into the cage that at least one pigeon had escaped. Which of the following does not represent a possible (x,y) pair?

- A (10,8)
- B (7,2)
- C (25,6)
- D (12,4)

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Question 9

In a hospital there were 200 diabetes, 150 hyperglycaemia and 150 gastro-enteritis patients. Of these, 80 patients were treated for both diabetes and hyperglycaemia. Sixty patients were treated for gastro-enteritis and hyperglycaemia, while 70 were treated for diabetes and gastroenteritis. Some of these patients have all the three diseases. Dr. Dennis treats patients with only gastro-enteritis. Dr. Paul is a generalist. Therefore, he can treat patients with multiple diseases. Patients always prefer a specialist for their disease. If Dr. Dennis had 80 patients, then the other three doctors can be arranged in terms of the number of patients treated as:

- A Paul > Gerard > Hormis
- B Paul > Hormis > Gerard
- C Gerard > Paul > Hormis
- D Cannot be determined

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Question 10

Three labeled boxes containing red and white cricket balls are all mislabeled. It is known that one of the boxes contains only white balls and one only red balls. The third contains a mixture of red and white balls. You are required to correctly label the boxes with the labels red, white and red and white by picking a sample of one ball from only one box. What is the label on the box you should sample?

- A White
- B Red
- C Red and White
- D Not possible to determine from a sample of one ball

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Instructions

A boy is asked to put one mango in a basket when ordered 'One', one orange when ordered 'Two', one apple when ordered 'Three', and is asked to take out from the basket one mango and an orange when ordered 'Four'.

A sequence of orders is given as: 1 2 3 3 2 1 4 2 3 1 4 2 2 3 3 1 4 1 1 3 2 3 4

Question 11

How many total oranges were in the basket at the end of the above sequence?

- A 1
- B 4
- C 3
- D 2

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Question 12

How many total fruits will be in the basket at the end of the above order sequence?

- A 9
- B 8
- C 11
- D 10

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CAT Syllabus PDF



Instructions

Answer the questions based on the following information. The following operations are defined for real numbers.

$a \# b = a + b$, if a and b both are positive else $a \# b = 1$

$a \nabla b = (a \times b)^{a+b}$ if $a \times b$ is positive else $a \nabla b = 1$.

Question 13

$$\frac{((1 \# 1) \# 2) - (10^{1.3} \nabla \log_{10} 0.1)}{1 \nabla 2}$$

- A $\frac{3}{8}$
- B $\frac{4 \cdot \log_{10} 0.1}{8}$
- C $\frac{(4 + 10^{13})}{8}$
- D None of these

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Question 14

$$\frac{(2\#1)}{(1\triangledown2)} =$$

A $\frac{1}{8}$

B 1

C $\frac{3}{8}$

D 3

 [VIEW SOLUTION](#)



Answers

1. **C** 2. **B** 3. **B** 4. **B** 5. **D** 6. **D** 7. **A** 8. **A**
9. **D** 10. **C** 11. **D** 12. **C** 13. **A** 14. **C**

Explanations

1. **C**

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co-authored with A and C. Since F has min Erdos number, the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians = 24

Erdos number at the third day: $f+1,b,f+1,d,e,f,g,h$

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians = 20 So here the difference of 4 [24-20] arose, which means e will be $f+5$ initially and changed to $f+1$ after co-authoring with F.

So the Erdos number at the third day: $f+1,b,f+1,d,f+5,f,g,h$

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be $f+5$ because then there will be two mathematicians with the same Erdos number $f+1$.

So five mathematicians will have $f+1$, one with $f+5$, one with f , one with some different value say x

$$5(f+1) + f + 5 + f + x = 24$$

$$7f + x = 14$$

The only value which satisfies the above equation is $f=1, x=7$

Erdos number at the end of fifth day, $f+1,b,f+1,d,f+1,f,g,h$ On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	$f+1$	$f+1$
B	b	b	b
C	c	$f+1$	$f+1$
D	d	d	d
E	e	$e = f+5$	$f+1$
F	$f(\text{min})$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Hence erdos no. of E at the beginning of conference would be $f+5 = 6$.

 VIDEO SOLUTION

2. **B**

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	f+1	f+1
B	b	b	b
C	c	f+1	f+1
D	d	d	d
E	e	e = f+5	f+1
F	f(min)	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Since at the end of the 3rd day 5 people had identical erdos no.(f+1) so : $5*(f+1) + f + f + 5 + x = 24$; Only $f=1$ and $x = 7$ satisfies the equation. So out of 5 people who had identical erdos no. at the end of day 3, 2 of them had different nos. at the beginning. So there were $5-2 = 3$ participants who had the same Erdos number at the beginning of the conference.

VIDEO SOLUTION

3. B

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number ,the values of c,a will change to f+1 and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day:f+1,b,f+1,d,e,f,g,h

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians=20 So here the difference of $4[24-20]$ arose, which means e will be f+5 initially and changed to f+1 after co-authoring with F.

So the Erdos number at the third day:f+1,b,f+1,d,f+5,f,g,h

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be f+5 because then there will be two mathematicians with the same Erdos number f+1.

So five mathematicians will have f+1, one with f+5, one with f, one with some different value say x

$$5(f+1)+f+5+f+x=24$$

$$7f+x=14$$

The only value which satisfies the above equation is $f=1, x=7$

Erdos number at the end of fifth day,f+1,b,f+1,d,f+1,f,g,h On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	f+1	f+1
B	b	b	b
C	c	f+1	f+1
D	d	d	d
E	e	e = f+5	f+1
F	f(min)	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Erdos no. of C at the end is $f+1 = 1+1 = 2$. Hence option B.

4. B

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number, the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos value then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3

Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day: $f+1,b,f+1,d,e,f,g,h$

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged.

Sum of the Erdos numbers of eight mathematicians=20

So here the difference of $4[24-20]$ arose, which means e will be $f+5$ initially and changed to $f+1$ after co-authoring with F.

So the Erdos number at the third day: $f+1,b,f+1,d,f+5,f,g,h$

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be $f+5$ because then there will be two mathematicians with the same Erdos number $f+1$.

So five mathematicians will have $f+1$, one with $f+5$, one with f , one with some different value say x

$$5(f+1)+f+5+f+x=24$$

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The only value which satisfies the above equation is $f=1,x=7$

Erdos number at the end of fifth day, $f+1,b,f+1,d,f+1,f,g,h$

On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
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F	$f(\text{min})$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Hence the person having the largest Erdos number at the end of the conference must have had Erdos number 7 . Hence option B.

5. D

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number, the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day: $f+1, b, f+1, d, e, f, g, h$

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians=20 So here the difference of 4 [24-20] arose, which means e will be $f+5$ initially and changed to $f+1$ after co-authoring with F.

So the Erdos number at the third day: $f+1, b, f+1, d, f+5, f, g, h$

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The only value which satisfies the above equation is $f=1, x=7$

Erdos number at the end of fifth day, $f+1, b, f+1, d, f+1, f, g, h$ On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	$f+1$	$f+1$
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C	c	$f+1$	$f+1$
D	d	d	d
E	e	$e = f+5$	$f+1$
F	$f(\min)$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

So B,D ,F,G,H are 5 participants in the conference who did not change their Erdos number during the conference.

 VIDEO SOLUTION



6. D

Considering option,

In 1st we get product of a and b which is not required In 2nd we get fraction 1.5 which doesn't help in finding average In 3rd we get $(a+b)/4$ after 2 operations which doesn't give average in further operations. only in 4th option we get overall and exact average of A, B and C .

$$/(X(@ (/ (@ (B, A), 2), C), 3), 2) = /(X(@ (/ (A+B)/2), 2), C), 3), 2)$$

$$= /(X(@ (A+B), C), 3), 2) = /(X((A+B, C)/2, 3), 2)$$

$$= /[(A+B+C)/6, 2]$$

$$= (A+B+C)/3$$

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7. A

Considering 1st option $/(@ (A, B), 2)$ First operation is average of A and B so we get $(a+b)/2$ and then we in next operation we multiply it by 2. So we get addition of A and B . So option A

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8. **A**

If the number of mynahs is 7 or more and if 7 birds escaped, there is a possibility that no pigeon escaped. But the bird-keeper was able to conclude that atleast one pigeon escaped. So, the number of mynahs is less than 7.

From the options, we can see that only option a) violates this condition.

[VIEW SOLUTION](#)

9. **D**

We don't know out of other 2 doctors which doctor is specialist in which disease.

So it is not possible to find out the exact order.

[VIEW SOLUTION](#)

10. **C**

All of them can be mislabeled in 2 ways:

Red box - white label

White box - Red and white label

Red and white box - Red label

Red box - red and white label

White box - Red label

Red and white box - White label

So, we would try the box with the red and white label and if it has a white ball, labelling to the boxes is done as per case 1. If it has a red ball labelling is done as per case 2.

Note: It's not a good idea to try the white label as if we get a red ball, we can't make out if we are picking from red box or red and white box. Similarly if we try from the box with red label and we get a white ball, again we can't make out if it is coming from the white box or the red and white box.

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11. **D**

First counting only nos. 1,2 and 3.

So, we have 6 mangoes, 6 oranges and 7 apples.

We have 4 times 4 so finally we have 2 mangoes, 2 oranges and 7 apples, which is a total of 11 fruits.

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12. **C**

On counting only numbers 1,2 and 3, we have 6 mangoes, 6 oranges and 7 apples.

We have 4 times number 4 => Finally we have 2 mangoes, 2 oranges and 7 apples. So, a total of 11 fruits.

[VIEW SOLUTION](#)

13. **A**

$$((1\#1)\#2) = 2\#2 = 4$$

$$\log_{10} 0.1 = \log_{10} 10^{-1} = -1$$

As $(10^{1.3} \times -1)$ is not positive, $(10^{1.3} \nabla \log_{10} 0.1) = 1$

$$1 \nabla 2 = (1 \times 2)^{(1+2)} = 2^3 = 8$$

Hence, the fraction is $(4-1)/8 = 3/8$

[VIEW SOLUTION](#)

14. **C**

$$2\#1 = 2 + 1 = 3$$

$$1 \nabla 2 = (1 \times 2)^{(1+2)} = 2^3$$

So answer will be $\frac{3}{8}$

[VIEW SOLUTION](#)

